

Votrax®

A Division of Federal Screw Works
500 Stephenson Highway
Troy, Michigan 48084

SC-01 SPEECH SYNTHESIZER DATA SHEET

Votrax® CMOS Phoneme Speech Synthesizer

GENERAL DESCRIPTION

The SC-01 Speech Synthesizer is a completely self-contained solid state device. This single chip phonetically synthesizes continuous speech, of unlimited vocabulary, from low data rate inputs. Figure 1.

Speech is synthesized by combining phonemes (the building blocks of speech) in the appropriate sequence. The SC-01 Speech Synthesizer contains 64 different phonemes which are accessed by a 6-bit code. It is the proper sequential combination of these phoneme codes that creates continuous speech.

The SC-01 Speech Synthesizer is cost-effective, consumes minimal power and enables in-house product development without vendor dependency. Signals from the SC-01 are applied to an audio output device to amplify and distribute the synthesized speech. See Figure 2.

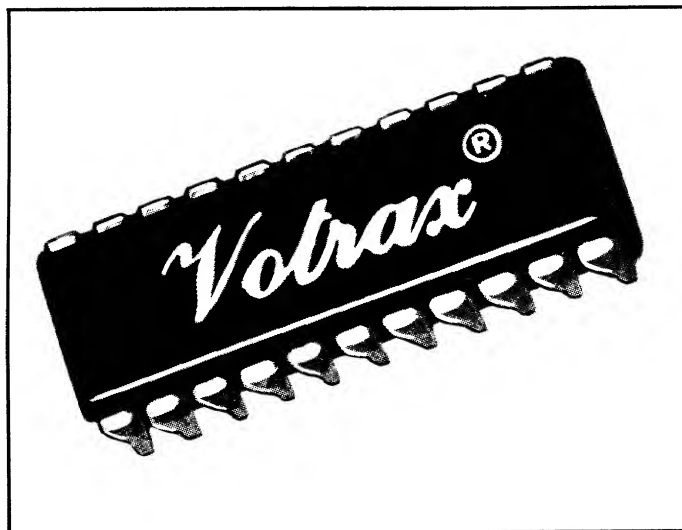


Figure 1. Votrax® SC-01 Speech Synthesizer

FEATURES

- Single CMOS chip
- 70 bits per second
- 22 pin package
- 9 ma. current drain
- Wide voltage supply range
- Latched 5v. compatible inputs
- Digital pitch level inputs
- Automatic inflection
- On-chip master clock circuit
- Optional external master clock
- Variety of voice effects
- Sound effects
- Customer product security

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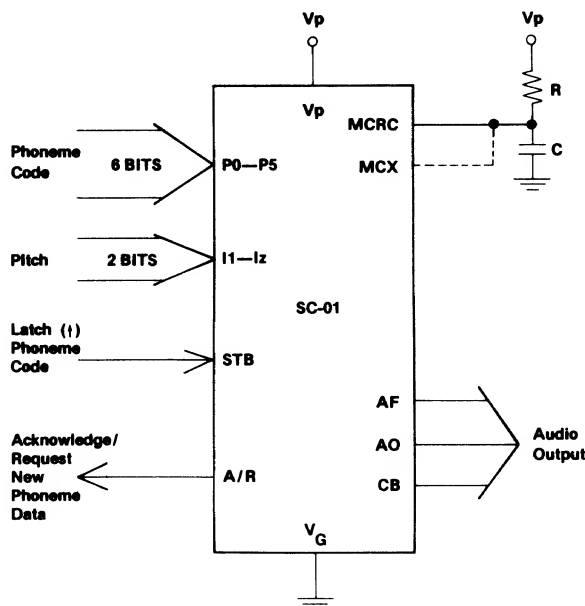


Figure 2. SC-01 Flow Diagram

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PHYSICAL DESCRIPTION

The SC-01 Speech Synthesizer comprises a 22 pin monolithic (single level or substrate) integrated circuit of CMOS (Complementary Metal On Silicon) design. See Figure 3. The P and N areas of the substrate, made by ion implantation, create a push-pull (Class B) transistor amplification and digital signal processing network. High impedance, rapid signal processing, and minimal current drain result from this technology.

ELECTRICAL DESCRIPTION

The SC-01 Speech Synthesizer is a program-compatible with existing Votrax[®] phoneme synthesizers. It requires 70 bits of data per second for continuous speech production. The 6-bit phoneme codes are 5 volt compatible and are latched for data bus line applications. A phoneme-construction algorithm and filters, within the chip, create the synthesized audio output.

PHONEME DESCRIPTION

Table 1 lists the 64 phonemes produced by the SC-01. Each phoneme code is accompanied by its symbol, average duration time, and an example. The underlined segments of the example word demonstrate the phoneme use, i.e., sound to be pronounced.

Table 2 subdivides the 64 phoneme symbols into seven categories. Each category represents a different production feature. The first six categories are characterized by voiced, fricative (expired voice), and nasal sounds. The seventh category is characterized by phonemes with no sound output.

PHONEME PROGRAMMING

Manual Operations: Votrax[®] maintains a library of phonetically programmed words. Reference to this library and programming manuals will aid in word synthesis.

Automatic Operations: Votrax[®] can supply a micro-computer system for automatic conversion of English text into phoneme sequences. This system is particularly useful for in-house vocabulary development and product security. Contact Votrax[®] for further information.

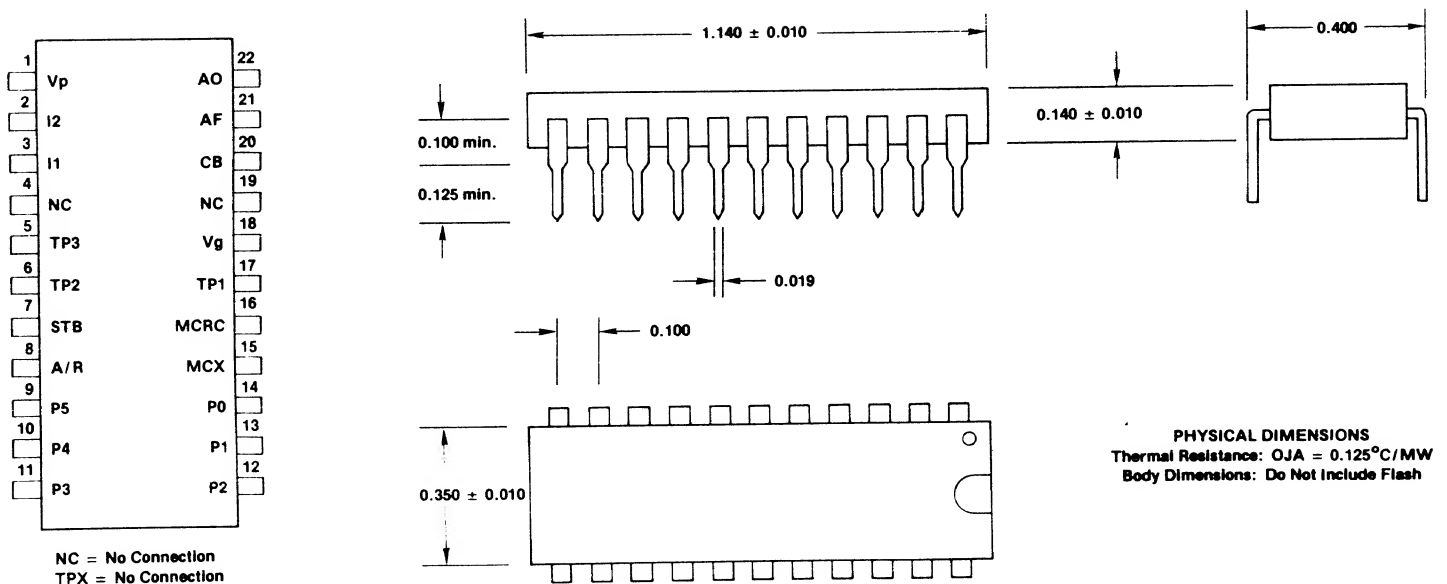


Figure 3. SC-01 Footprint and Outline Dimensions

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Table 1. Phoneme Chart

Phoneme Code	Phoneme Symbol	Duration (ms)	Example Word
ØØ	EH3	59	jacket
Ø1	EH2	71	enlist
Ø2	EH1	121	heavy
Ø3	PAØ	47	no sound
Ø4	DT	47	butter
Ø5	A2	71	made
Ø6	A1	103	made
Ø7	ZH	90	azure
Ø8	AH2	71	honest
Ø9	I3	55	inhibit
ØA	I2	80	inhibit
ØB	I1	121	inhibit
ØC	M	103	mat
ØD	N	80	sun
ØE	B	71	bag
ØF	V	71	van
1Ø	CH*	71	chip
11	SH	121	shop
12	Z	71	zoo
13	AW1	146	lawful
14	NG	121	thing
15	AH1	146	father
16	OO1	103	looking
17	OO	185	book
18	L	103	land
19	K	80	trick
1A	J*	47	judge
1B	H	71	hello
1C	G	71	get
1D	F	103	fast
1E	D	55	paid
1F	S	90	pass

Phoneme Code	Phoneme Symbol	Duration (ms)	Example Word
2Ø	A	185	day
21	AY	65	day
22	Y1	80	yard
23	UH3	47	mission
24	AH	250	mop
25	P	103	past
26	O	185	cold
27	I	185	pin
28	U	185	move
29	Y	103	any
2A	T	71	tap
2B	R	90	red
2C	E	185	meet
2D	W	80	win
2E	AE	185	dad
2F	AE1	103	after
3Ø	AW2	90	salty
31	UH2	71	about
32	UH1	103	uncle
33	UH	185	cup
34	O2	80	for
35	O1	121	aboard
36	IU	59	you
37	U1	90	you
38	THV	80	the
39	TH	71	thin
3A	ER	146	bird
3B	EH	185	get
3C	E1	121	be
3D	AW	250	call
3E	PA1	185	no sound
3F	STOP	47	no sound

* T must precede CH to produce J sound.

D must precede J to produce CH sound.

Table 2. Phoneme Categories According to Production Features

Voiced					'Voiced' Fricat.	'Voiced' Stop	Fricat. Stop	Frica-tive	Nasal	No Sound
E	EH	AE	UH	OO1	Z	B	T	S	M	PAØ
E1	EH1	AE1	UH1	R	ZH	D	DT	SH	N	PA1
Y	EH2	AH	UH2	ER	J	G	K	CH	NG	STOP
Y1	EH3	AH1	UH3	L	V		P	TH		
I	A	AH2	O	IU	THV			F		
I1	A1	AW	O1	U				H		
I2	A2	AW1	O2	U1						
I3	AY	AW2	OO	W						

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Phoneme 6-Bit Selection Code (P0-P5): Data input is to six pins. Latching is controlled by the strobe (STB) signal.

Strobe (STB): Latching occurs on rising edge of strobe signal.

Inflection Level Setting (I1, I2): Instantaneously set pitch of voiced phonemes.

Acknowledge/Request (A/R): Acknowledges receipt of phoneme data (signal goes from high to low one master clock cycle following active edge of STB signal). Also indicates timing out of old phoneme concurrent with request for new phoneme data (signal goes from low to high).

NOTE

If external phoneme timing is desired, phoneme requests can be ignored. However, best speech is realized with internal timing.

Master Clock Resistor-Capacitor (MCRC): This input determines the internal master clock frequency. Select R-C values for 720 KHz to achieve standard phoneme timing. Connect this input to MCX when using internal clock; ground when using external clock.

Varying clock frequency varies voice and sound effects. As clock frequency decreases, audio frequency decreases and phoneme timing lengthens. Figures 6 and 7 illustrate manual and DAC (Digital to Analog Converter) voice variation schematics, respectively.

Master Clock External (MCX): Allows control by an external clock signal.

NOTE

Ground MCRC during MCX operation.

Audio Output (AO): Supplies analog signal to audio output device.

Audio Feedback (AF): Used with Class A or Class B transistor audio amplifiers for added stability.

Class B (CB): Current source for Class B transistor audio amplifier.

Table 3. Timing Specifications

CHARACTERISTIC	SYMBOL	MIN	TYP	MAX	UNIT
Input Setup Time (P_I to STB)	T_S			450	NS
Input Hold Time (P_I to STB)	T_H			0	NS
Rise Time of STB Edge (.8v to 4v)	T_{RS}			100	NS
A/R Width (A/R Connected to STB) ⁺	T_{ARW}	1	1.3	2	μs
STB Width	T_{SW}	200			NS
STB Low*	T_{SL}	*			
Propagation Delay (STB to A/R after 2 μs)	T_{DAR}			500	NS
A/R Rise Time (Capacitive load = 30pf)	T_{RAR}			100	NS
A/R Fall Time (Capacitive load = 30pf)	T_{FAR}			100	NS
Time from A/R Request to STB Service)	T_{ARS}	0	500		μs
Time of Phoneme Duration ⁺	T_{PH}	47	107	250	MS

⁺ Dependent on Master Clock frequency: 720KHz = 1.3 μs

* Strobe must remain low (64 x master clock period) before rising edge.

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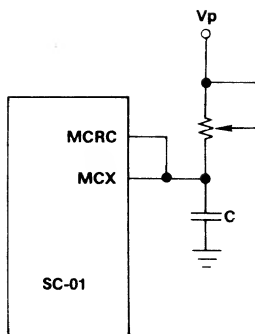


Figure 6. Variable Voice by Potentiometer Control

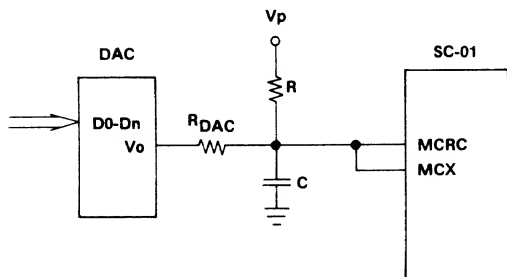


Figure 7. Variable Voice by DAC Current Injection

TYPICAL APPLICATIONS

General: The SC-01 Speech Synthesizer is easily designed into systems ranging in complexity from ROM/counters to microprocessor controllers.

Single Message System: See Figure 8. When the counter is released (START is TRUE), the message is clocked out of the ROM by the A/R signal. The system must be stopped when DONE is TRUE.

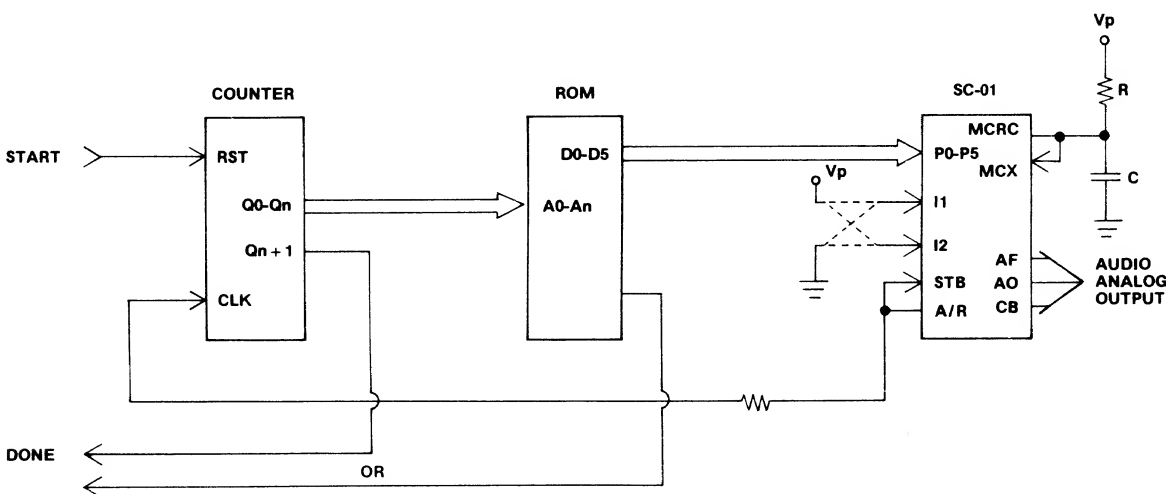


Figure 8. Single Message System

NOTE

Data at address 0 must be a pause phoneme code.

Multiple Message, Fixed Block Size: See Figure 9. Message address block is loaded into the counter. The message is then clocked out of the ROM by the A/R signal.

NOTE

Message Block = 2^n maximum.

Multiple Message, Variable Block Size: See Figure 10. The microprocessor loads phonemes into a data bus. The A/R signal generates an interrupt request for each new phoneme.

CONNECTING THE AUDIO OUTPUT DEVICE

Audio Output: The AO signal has a maximum voltage swing of 1.5 volts below V_p or above V_G , depending upon the phoneme selected. Furthermore, the AO signal is D.C. biased by a factor of $V_p/2$.

NOTE

If additional audio amplifier stability is not needed, connect the AO pin to the AF pin.

Class A Amplifier: See Figure 11. For a single transistor amplifier, the selection of R, C, or R_s values depends upon the value of V_p and the desired audio level.

NOTE

The CB pin is not used for Class A amplifiers.

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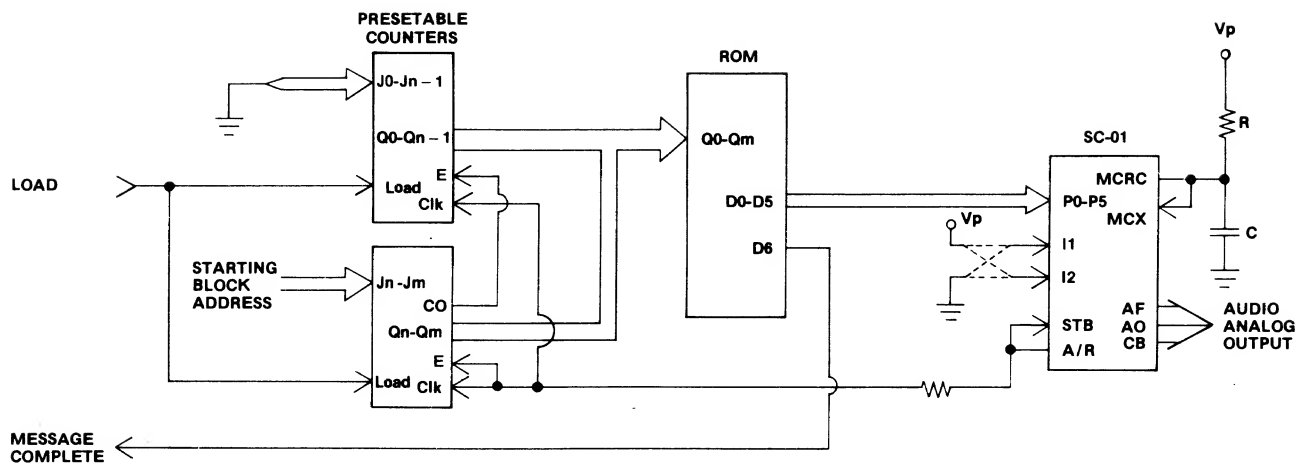


Figure 9. Multiple Message, Fixed Block Size

Class B Amplifier: See Figure 12. A current source (CB) is required for this push-pull amplifier.

NOTE

Minimum power is consumed when speech is inactive. When $V_p = +12.0$ volts and $R_s = 40$ ohms, the bias current drain is approximately 3.5 milliamps.

Controlling Audio Output Power: See Figure 13. A resistor or potentiometer from the speaker to ground can be used to control the audio output power.

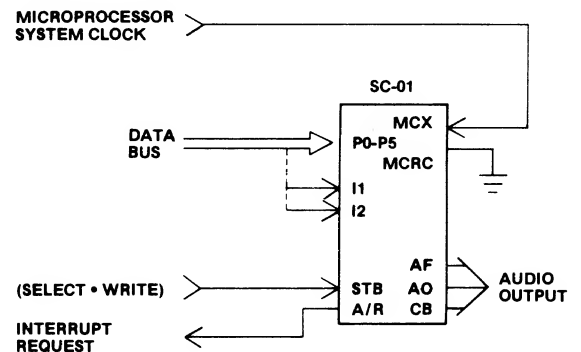


Figure 10. Multiple Message, Variable Block Size

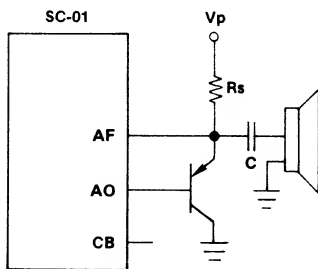


Figure 11. Class A Amplifier

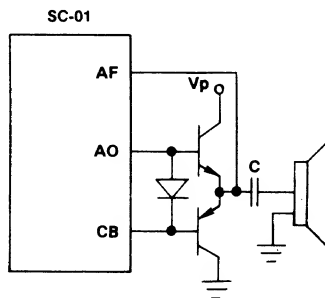


Figure 12. Class B Amplifier

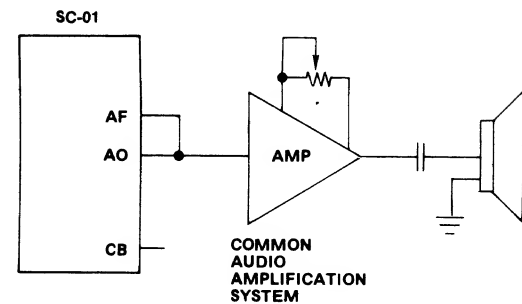


Figure 13. Controlling Audio Output Power

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Table 4. Analog Output Specifications

CHARACTERISTIC	MIN	MAX	UNIT
Output Voltage (AW Phoneme)	.39·V _p	.58·V _p	V _{p-p}
Output Bias Current ** (.4v CB V _p)	3.5	7.3	MA

ELECTRICAL CHARACTERISTICS: T_o = 0 to 70°C, V_p = 7 to 14 V_{DC}

CHARACTERISTIC	MIN	TYP	MAX	UNIT
Digital Input Impedance	1 meg.			OHM
Input Capacitance (P ₁ , STB)			3	pf
Input Capacitance (I1, I2, MCX)			8	pf
Digital Input Logic "0" (except I1, I2, MCX)	V _G +0.8		V _G -0.5	V _{DC}
Digital Input Logic "0" (I1, I2, MCX)	V _G +1.0			V _{DC}
Digital Input Logic "1" (except I1, I2, MCX)	V _p +0.5		V _G +4.0	V _{DC}
Digital Input Logic "1" (I1, I2)			V _p -1.0	V _{DC}
Digital Input Logic "1" (MCX)			4.6	V _{DC}
Digital Output Logic "0" (I sink = 0.8MA)			V _G +0.5	V _{DC}
Digital Output Logic "1" (I source = 0.5MA)	V _p -0.5			V _{DC}
Power Supply Current V _p = 9v		9.1		MA
V _p = 9v **		11	18	MA
V _p = 14v **		18	27	MA
Master Clock Frequency		720K		Hz
Master Clock Resistor Value (MCRC)	6.5K			OHM
Master Clock Capacitor Value (MCRC)			300	pf

** With CB, AF, AO connector for Class B audio amplifier (see APPLICATION NOTES)

Note: TP1, TP2, TP3 must be left open for normal operation.

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Table 5. Absolute Maximum Ratings

ABSOLUTE MAXIMUM RATINGS *

RATING	SYMBOL	VALUE	UNIT
Power Supply Voltage	V_p	20	V_{DC}
Power Dissipation at 25°C	P_{DM}	650	MW
Derating Above 25°C		5	MW
Operating Ambient Temperature	T_o	0 to 70	°C
Storage Temperature	T_{STG}	-55 to 125	°C
Input Voltage	V_{INM}	-0.5 to $V_p+0.5$	V_{DC}
DC Current Max. Above $V_p+0.5v$	I_{INM}	1.0	MA
Lead Temperature (soldering 10 sec.)	T_L	300	°C

* Operation above these limits could damage the device.

NORMAL OPERATING CONDITIONS: $V_v = V_p = 14v$, $0^\circ C = T_o = 70^\circ C$

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